AI ENGINEERING PORTFOLIO

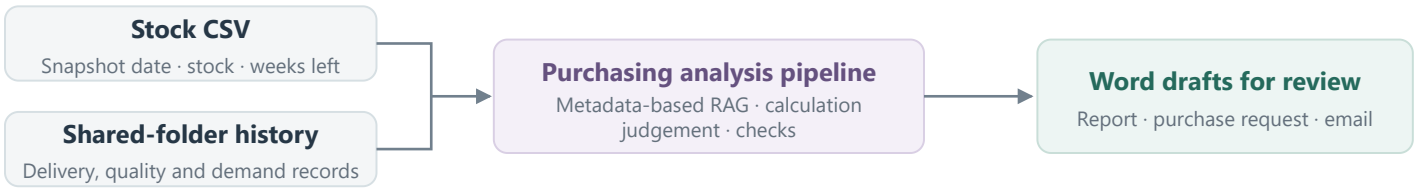
Multi-agent Purchasing AI Suite

A purchasing-decision support system

Analyses a stock CSV together with the supplier and item history
and writes **a purchasing analysis report, a purchase request and a supplier email draft**.

Soomin Myung · AI Engineer · planning, design, development and operation

> It **combines the stock data with the accumulated item and supplier history**, analysing delivery, quality and demand changes together, to support purchasing decisions with context and evidence.



Code and LLM roles	History retrieval	Document checks	A model for each task	Time and cost
Code calculates, LLM judges Code: figures, retrieval, assembly LLM: reading history, writing	**Metadata-based RAG** Supplier, item and date headers find only the relevant history	**36 checks** Code and LLM judge Rewrites with feedback (up to 2)	**Model/effort benchmarks** Same inputs → scored by checks → manual review (HITL)	**About 80 s · $0.05** One supplier, seven items

01 Background

At Korea Foods, a food-distribution company with annual revenue of about £89M (year ending June 2025), the **purchasing team** monitored stock and placed orders for tens of thousands of items and was **always short of time.** The supplier and product history, well organised under supplier, item and date headers, could not be searched and analysed fully for each order decision, which could **weaken the company's position in negotiations**. The system **uses AI's speed at finding and reading documents** to combine this history with the stock data. Finding and reading the history, analysing it and writing the documents could take a buyer hours; it now takes **about 80 seconds per supplier**, with ready-to-use **drafts of an analysis report, a purchase request and a supplier email**.

02 Documents for each purchasing stage and role

The buyer runs an SAP user query on the item codes and exports the result as a CSV (item code, item name, supplier, available stock, weeks to stockout); **uploading it and starting a run returns each document as a Word draft**. **Separate documents are written for the buyer, the approvers and the supplier**. Ordering, negotiation and enquiries happen per supplier, so items are **grouped by supplier for the analysis and the documents**.

01 / BUYER

Purchasing analysis report

Lead-time and quantity changes citing the history, with risk and order dates recalculated; questions to settle internally or ask the supplier; actions with deadlines.

02 / APPROVERS

Purchase request (PR)

Purpose, justification and item table. A temporary note at the top lists open questions and actions. Resolve them, update the document and remove the note.

03 / SUPPLIER

Email draft

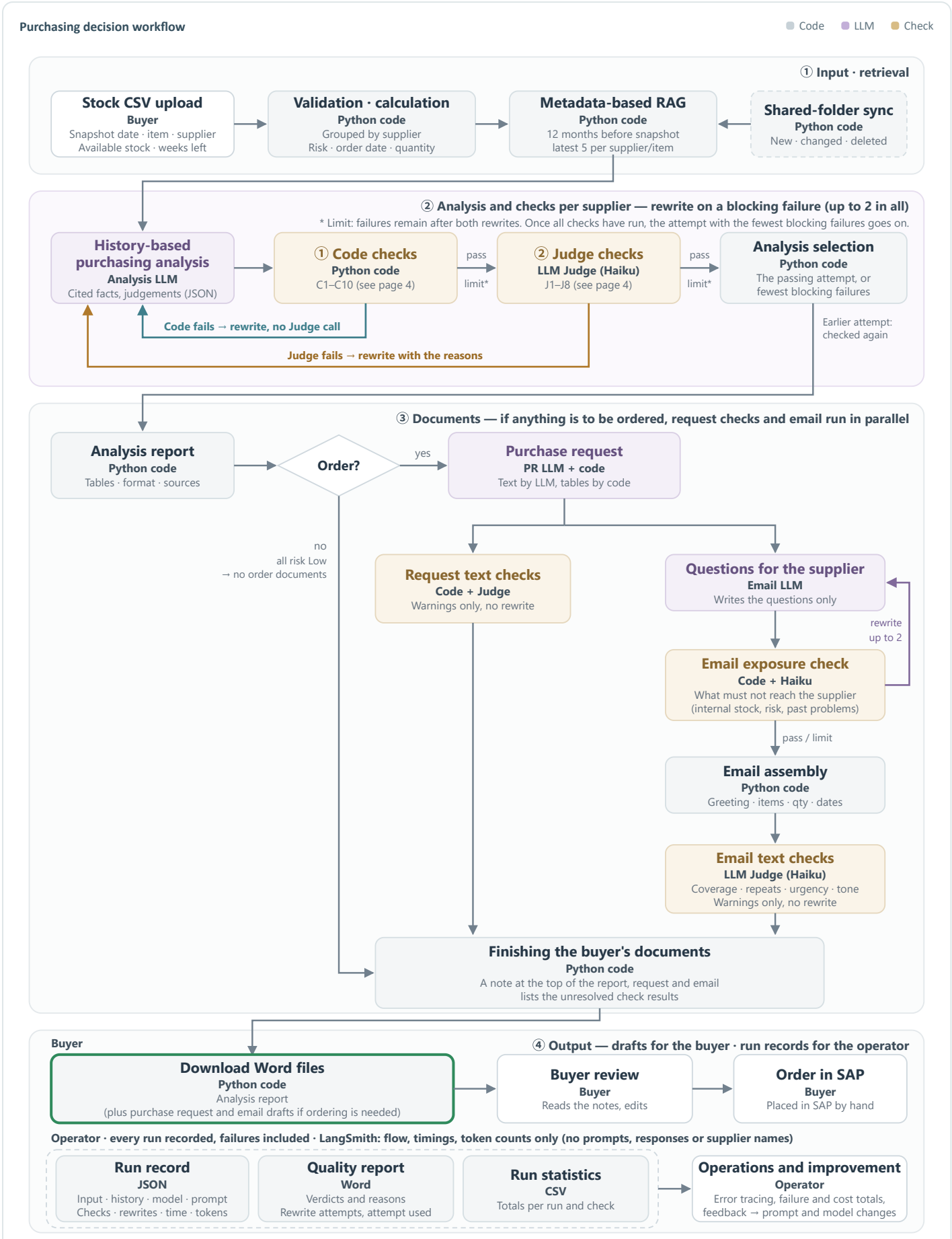
Quantities, requested delivery and points to confirm, checked so that no internal stock, risk or analysis leaks out.

If every item's stock risk is Low, the purchase request and email are skipped. The buyer reviews the drafts and places the order in SAP (pipeline and checks: pages 2–4).

Built on the system developed and piloted at Korea Foods; the architecture, evaluation and models were developed further as a personal project.

The pipeline from input to documents

Input → analysis and checks → documents written and checked in parallel → drafts for the buyer, run records for the operator.



Evidence traceability and output quality checks

Code does the arithmetic and the document assembly; the **LLM** makes the history-based purchasing judgements and writes for each reader.

01 Division of roles and the evidence trail

By	Role
Code	Input checks; risk, quantity and order-date calculation; history retrieval; assembly of tables, formatting and sources
LLM	Facts from the history; lead-time and quantity changes; questions and recommended actions; the prose

Evidence trail: source documents [1], [2]… → facts [F1], [F2]… → changes, questions and recommended actions. Facts cite their documents, and changes, questions and actions cite their facts; facts from the stock figures have no document number and are listed apart. Code checks the references and links; the LLM judge checks whether the cited evidence logically supports the judgements.

Stock risk levels · W = weeks to stockout, L = lead time (18 weeks by default)
W<L: Critical · W<L+8: High · W<L+17: Medium · otherwise Low, applied in that order.

01 / Metadata-based RAG

Searched by the supplier, item, code and date headers: the latest five documents per supplier and item from the 12 months before the snapshot. Full source documents are provided without chunking; the shared folder is kept in sync.

02 / Consistent figures

Lead time 18 weeks by default; the suggested quantity covers 26 weeks of sales after arrival. When the LLM changes a lead time, **code recalculates the risk, order date and projected stockout duration** and checks the LLM's figures.

03 / Email guardrails

Internal stock, risk and questions not meant for the supplier are **kept out of the LLM's input**; quantities and dates come from code. Code and a separate LLM **check for exposure** of internal data, past supplier problems and other items.

02 Using the check results and run records

Quality report: the check verdicts and reasons for the analysis, purchase request and email, and the result of each analysis rewrite, with the attempt used for the documents. The report is kept on the server for the operator.

Review notes: code puts each document's unresolved check results in a temporary note at the top of the analysis report, purchase request and email drafts. Unresolved analysis-check results are also shown on the purchase request, since it is based on the analysis. Even when every item is Low and only the report goes out, the buyer can see any unresolved checks affecting the conclusion.

An analysis that still fails: a blocking failure sends the analysis back for a rewrite, limited to two so that time and cost cannot grow without end. A rewrite can also be worse, so if both fail, the documents use the attempt with the fewest blocking failures. That attempt may have skipped the judge, so it is checked again by code and the judge, and the report and notes are updated.

Run records (JSON, CSV): apart from the quality report, **every run writes a detailed JSON record** on the company server, failed runs included: the CSV as received, the history documents the model read, model and prompt versions, every output and check result, and each step's time, tokens and errors. **Run and check statistics accumulate in CSV files** to track failures, rewrites and costs by version. Company data stays in these records only.

LangSmith: pipeline tests and prompt comparisons on synthetic data in development; time, token, cost and failure monitoring in production. Only the run tree, timings, token counts and model names are sent: prompt and response text is hidden, and supplier names in run names and metadata are removed in code.

Deployment: piloted at Korea Foods on the company server with the GPT-4o API. Public demo CI/CD with Docker, GitHub Actions and Cloud Run (currently inactive).

MCP: history and calculation servers built and tested but not used in production. Code fixes the order of retrieval and calculation, so the model cannot skip a tool call and leave out history or change the evaluation input.

Quality checks for each document

Each document has its own checks: **code** checks format, figures and citations; an **LLM** checks meaning, evidence and exposure.

01 Analysis checks

ANALYSIS QUALITY CHECKLIST

The analysis LLM's **lead-time and quantity changes, questions and recommended actions** are checked by code and by a separate LLM judge.

Blocking: a failure triggers a rewrite with feedback, up to two. Errors left are shown in the note at the top of the draft and in the operator's quality report.

Review: no rewrite; the note at the top asks the buyer to check.

Code — format and figures

10 CHECKS

C1	Block	Are the required fields and types as specified?
C2	Block	Does it mention only items in the input?
C3	Block	Does the original quantity match the code's recommendation?
C4	Block	Does each fact cite a source for its supplier or items, and each judgement cite facts for the relevant supplier or items?
C5	Block	Are lead-time changes valid and revised quantities calculated according to the rules?
C6	Block	After changes, does each Critical item have an action citing a finding about its stock?
C7	Review	Does the summary name every Critical item?
C8	Review	Is the summary within 100 words?
C9	Block	Does each 'before → after' pair in the text (e.g. '18 → 22 weeks') match a proposed change?
C10	Block	Do the model's figures after changes match code's?

LLM Judge — evidence and judgement

14 CHECKS

J1a	Block	Is the stated history actually in the documents?
J1b	Block	Is the history neither exaggerated nor understated?
J1c	Block	Are historical facts attributed to the correct supplier and items?
J2a	Review	Is there a question on the supplier issues in the history?
J2b	Review	Is there a question on the item issues in the history?
J2c	Review	Do the questions avoid assuming unsupported problems?
J3a	Block	Is each change in the direction its evidence supports?
J3b	Block	Is each change's size in keeping with the evidence?*
J3c	Block	Are changes applied to all items covered by the same rationale, unless an exception is justified?
J4	Block	Do stock figures in the text match code's final figures?
J5	Block	Are questions for the supplier free of internal information?
J6	Review	Do the cited facts logically support the questions and actions?
J7	Review	Have all documents on issues affecting this order been used?
J8	Review	Are findings free of purchasing judgement or interpretation?

With nothing to check (no change, no supplier question), a check passes or is skipped, depending on the check. Each verdict is recorded with its reason, and the same checks serve model comparison, checking during runs and the selection of SFT training answers.

* J3b: where the evidence gives no size, an estimate passes only if it is in keeping with what the evidence says and is labelled as an estimate to be confirmed.

02 Purchase request and email checks

PURCHASE REQUEST · SUPPLIER EMAIL

Purchase request: checks that **turning the analysis into text for the approvers neither distorts nor adds to the facts and figures**. No rewrite. Failed checks, including P3, are listed in a note at the top.

Email: it leaves the company, so exposure of internal information (E1–E3) is blocking. A failure triggers up to two rewrites with the reasons; any left are shown in the note at the top. Missing or repeated questions, urgent-item enquiries and tone (E4–E7) are review checks.

Purchase request — facts and figures kept

5 CHECKS

P1a · LLM	Review	Does it avoid adding facts or assessments not in the input?
P1b · LLM	Review	Are figures reproduced exactly, without evaluative wording?
P1c · LLM	Review	Where reworded, is the meaning unchanged?
P2 · Code	Review	Is the purpose 1–2 sentences long and the justification 2–4?
P3 · Code	Block	Does it cite only the history documents given?

Email — internal data and request quality

7 CHECKS

E1 · Code·LLM	Block	Does it avoid disclosing internal terms, stock, risk or analysis?
E2 · LLM	Block	Does it avoid mentioning past supplier problems?
E3 · Code·LLM	Block	Does it avoid mentioning items outside the order?
E4 · LLM	Review	Do the questions cover every point to confirm?
E5 · LLM	Review	Does it avoid repeating a question?
E6 · LLM	Review	Does it ask for the earliest dispatch date or faster delivery for urgent items?
E7 · LLM	Review	Is it a polite request, without repeating the item list or promising an order?

The same checks were used in the comparisons that chose the purchase-request and email models. E1–E3 are decided by the email validator (code + Claude Haiku), the other LLM checks by the LLM judge (Claude Haiku).

Comparing and choosing a model for each task

API model evaluation outputs

About 1,360

Analysis, purchase request and email comparisons, and more

API model check verdicts

About 38,600

Per-check verdicts from about 6,000 judge calls

LLM judge error detection

93.9%

51 samples (49 faulty, 2 sound), each judged 3 times

- **Candidates (GPT, Claude and Gemini models at several price tiers and reasoning effort settings) → run on the same inputs → scored on predefined checks (code and LLM judge: 24 for the analysis, 5 for the purchase request, 7 for the email) → read by the developer (HITL) → model chosen, prompt finalised.**
- Cost, latency and the differences in the actual wording were compared alongside the pass rates.
- A judge may favour outputs of its own model family, so in production the judge comes from a different family from the models it checks. With the judge on Claude (Haiku), and Gemini unreliable because of call errors, the purchase-request and email writers were compared among GPT models.

01 Analysis model — accurate figures and specific actions

Analysis agent comparison: 7 models, 9 settings, 3 rounds; the same 2 cases, 2–4 runs per setting. Time per analysis call; cost per run for one supplier, rewrites included, judge and documents excluded. Figures checked against code's recalculation after lead-time changes.

Model · effort	Call time	Analysis cost per run	Figures after changes	Manual review
`gpt-4o`	8–14 s	$0.039–0.064	Incorrect	Figure errors (7–8 of 20); extended the expected lead time based on a resolved customs delay.
`gpt-6-luna · medium`	21–36 s	$0.0015–0.0042	Correct	One supplier question; no specific deadlines in the actions.
`gpt-6-luna · high`	37–94 s	$0.0022–0.0042	Correct	No more specific than medium; over the 60 s response time limit 3 times.
`claude-sonnet-5-5 · high`	20–31 s	$0.039–0.109	Correct	No meaningful difference from medium, but slower and costlier.
`claude-sonnet-5-5 · medium`	10–22 s	$0.026–0.046	Correct	The most specific and wide-ranging questions; 1.3–1.6× Sol's cost.
`gemini-3.5-flash-lite`	—	—	—	Call failed: Google server overload (503).
`gemini-3.6-flash`	34 s	$0.031	Correct	503 errors in 3 of 4 runs; dropped for availability.
`gemini-3.8-flash`	—	—	—	2 of 3 test calls failed (503, 504); 1 took 169 s, over the time limit.
`gpt-6.1-sol · medium`	23–39 s	$0.021–0.033	Correct	Actionable follow-ups with explicit order and emergency delivery dates.

Sol medium chosen: specific purchasing actions with deadlines, at lower cost than claude-sonnet-5-5 · medium. Running the analysis on GPT and the judge on Claude (Haiku) reduces the risk of the judge favouring its own family. Gemini was dropped for server availability and stability: in three attempts over a week, 503 overloads and slow responses recurred every time.

02 LLM judge — catching every error, with consistent verdicts

Analysis and purchase-request judge comparison: detection and verdict changes on 51 samples (49 faulty, 2 sound), each judged 3 times; false alarms against 89 developer verdicts on 6 separate outputs. Set beforehand: detection ≥ 90%, false alarms ≤ 10%, verdict changes ≤ 5%.

Judge setting	Detection	Verdict changes	False alarms	Cost per verdict	Time
`gpt-6-luna`	89.1%	7.8%	17.0%	$0.0013	24 s
`gemini-3.1-flash-lite`	Rate-limit errors (429): 61% of verdicts missing. Detection on the samples answered: 40.4%.				
`claude-haiku-5-5 · forced tool, no reasoning`	83.7%	17.6%	2.7%	$0.0012	6 s
`claude-haiku-5-5 · medium`	91.2%	5.9%	9.5%	$0.0018	12 s
`claude-haiku-5-5 · high`	93.9%	3.9%	8.0%	$0.0024	16 s

Missed errors versus false alarms: a missed error stays in the document unmarked, and the buyer and approvers may set order quantities and dates on a wrong figure or fact. A false alarm appears in the review note and requires an extra check by the buyer. So **missing fewer errors came first, at the price of some false alarms.**

Haiku high chosen: the only setting to meet every criterion, with the highest detection. With a forced tool call fixing the verdict format, Haiku answered without reasoning, unlike luna, which reasons; once this was found and Haiku was made to reason before answering, detection rose from 83.7% to 93.9%.

> The purchase request and email restate the analysis, so they were compared less on reasoning than on **whether they adapt the style to the document's purpose without distorting or adding to the facts and figures**. Each setting ran on the same 25 inputs, and every output was manually reviewed.

03 Purchase request writer — keeping the facts and figures

Purchase request comparison: the purpose and justification paragraphs the approvers read. Code checks (sentence counts, citations), judge (added content, judgemental wording, exaggeration), manual review. gpt-4o and gpt-5.4-mini from round 2, the rest from the final round 5. Counts are outputs with a problem, out of 25.

Model · effort	Outputs with issues, manual review	Outputs with issues, judge	Time	Cost per request	Manual review
gpt-4o	22	19	2.8 s	$0.0047	Added a subjective supplier assessment not in the input ("proven track record").
gpt-5.4-mini · none	18	6	2.0 s	$0.0018	Exposed a system term from the prompt in the approval document ("pipeline").
gpt-5.4-mini · low	19	15	2.8 s	$0.0023	Called the order dates revised when nothing had changed ("revised order dates").
gpt-6-luna · none	7	2	3.6 s	$0.0002	Stated the order's arrival date earlier than it would actually be.
gpt-6-luna · low	6	3	4.1 s	$0.0003	Presented the company's inspection findings as the supplier's own report.
gpt-6-luna · medium	4	4	10.3 s	$0.0006	"Often occurs" weakened to "can occur".
gpt-6.1-sol · medium	0	0	12.8 s	$0.0076	No issues with style adaptation or preservation of facts and figures.
gpt-6.1-sol · low	1	1	5.3 s	$0.0056	One ambiguous date expression; otherwise no issues with style or factual accuracy.

Sol low chosen: in the final comparison with Sol medium, manual review found no problems in medium and only one ambiguous date in low, and low takes under half the time at about three quarters of the cost. gpt-5.4-mini, first on the sentence-count rule (24/25), was dropped after its outputs were read: a case the metrics alone could not decide.

04 Email writer — confidentiality and indirect questions about past issues

Email comparison: final round 3, the same 25 inputs; counts are emails with a problem, out of 25. Code check (E3, other items), judge (E1, E2, E4–E7: internal data, past problems*, missing or repeated points, dispatch date for urgent items, tone), manual review. The judge checked each email 3 times, and a check failing 2 or more times counts as failed. No setting exposed internal data (stock, risk, internal review), so the differences came from how past problems were asked about*.

Model · effort	Outputs with issues, manual review	Outputs with issues, judge	Time	Cost per email
gpt-5.4-mini · none	7	6	2.4 s	$0.0010
gpt-5.4-mini · low	6	6	2.5 s	$0.0013
gpt-6-luna · none	3	3	2.3 s	$0.0001
gpt-6-luna · low	1	2	3.4 s	$0.0002
gpt-6-luna · medium	0	2	5.2 s	$0.0003
gpt-6.1-sol · medium	0	0	5.0 s	$0.0033
gpt-6.1-sol · low	0	0	3.3 s	$0.0024

* On the same point (loose terminals in the January batch), gpt-5.4-mini asked it as it stood: “whether the loose-terminal issue in the January PowerModuleZ batch was isolated”; luna (none) left out the date and batch but still named the problem: “the cause of the loose-terminal failure”; Sol low made it a question about this order, without naming the past problem: “how terminal integrity will be verified for the items listed above”.

Sol low chosen: of Sol low and medium, the two with no problems in manual review or by the judge, it is faster (3.3 s) and asks without revealing the supplier's past problems. With one email per supplier, it costs about $0.21 more than luna per 100 suppliers.

05 Email validator — catching violations with few false alarms on compliant emails

Email validator comparison: detection measured on 13 real violations and on 64 test emails, copies of Sol low's 25 compliant emails each with one violation planted; false alarms on 162 compliant emails from 7 writer settings (25 of them by Sol low, the production model). Compliance is assessed against the seven email checks. Of Haiku medium's 8 false alarms, 7 were stricter readings of wording that hinted at a past supplier problem.

Validator	Real violations caught (13)	Planted violations caught (64)	False alarms, compliant emails (162)	False alarms, Sol low (25)	Time	Cost per check
gpt-4o-mini	13/13	61/64	55/162 (34.0%)	10/25 (40.0%)	0.8 s	$0.00005
claude-haiku-5-5 · low	12/13	61/64	47/162 (29.0%)	8/25 (32.0%)	1.9 s	$0.00014
claude-haiku-5-5 · high	11/13	59/64	8/162 (4.9%)	0/25 (0.0%)	2.4 s	$0.00021
claude-haiku-5-5 · medium	13/13	60/64	8/162 (4.9%)	0/25 (0.0%)	2.2 s	$0.00018

Haiku medium chosen: every candidate caught nearly every real violation, so the difference lay in false alarms, which send a compliant email back for a rewrite. gpt-4o-mini failed 55 of 162 compliant emails (34%), Haiku medium 8 (4.9%), and none of those written by Sol low. Haiku high had as many false alarms as medium but missed 2 violations, and was slower and costlier.

Self-hosting experiments and quality checks

Whether to replace the production model was judged on ① **serving performance** and ② **output quality**, measured separately.

01 Serving performance and output quality

① Serving: vLLM memory and throughput

One RTX 5090 · Gemma 4 12B. An infrastructure measurement, separate from model quality, so the untuned model was used, comparing bf16 and FP8 on real analysis requests.

Measure	bf16	FP8
Weights and runtime memory	23.9 GiB	13.3 GiB
Combined throughput, 16 requests	453.5 tok/s	834.5 tok/s
Cases passing all blocking checks	14–15/25	13/25

About **44% less memory**, and about **1.84×** the throughput at 16 concurrent requests.

The quality difference was within run-to-run variation (3 bf16 verdicts changed between runs; one FP8 run cannot establish a difference). The failures seen only in FP8 were mostly unmarked estimates (J3b): the history gave a reason for a longer lead time but not how much (e.g. a strike with no end date), and the model did not mark its +4 weeks as an estimate to confirm. The purchasing judgements themselves were sound.

② Output quality: before and after QLoRA SFT

Gemma 4 12B chosen from five open-weight models. 98 synthetic answers that passed the checks (85 training, 13 validation), and **6 QLoRA SFT runs: 25, 50 and 85 training examples × 2 seeds**.

Validation loss · 25→85 examples, seed mean **0.452 → 0.328**

Cases passing all blocking checks · before → after SFT **14/25 → 3–10/25**

What it learned: for an item that runs out before a new order can arrive, the supplier must be asked whether urgent or partial delivery is possible. In the 18 such cases, Sol asked this question 18 times and untuned Gemma 2 times; after training, Gemma asked it 19–21 times.

The model reused wording from internal actions in supplier questions, revealing **how many weeks our stock would be short**, e.g. "…CircuitBoardA **to bridge the 8-week gap**…". A supplier aware of our urgency gains leverage on price and allocation, so this is internal information, and J5 blocks it.

02 Tracing the SFT failure — internal-action wording in supplier questions

Comparing the saved outputs with the reasons for each verdict showed wording crossing from one field to another. The phrase "bridge the n-week stock gap", used in internal actions, went into the questions for the supplier as well.

Model · the 18 cases with an item running out before arrival	Supplier questions on urgent or partial delivery	Of which revealing the stock gap
`Production model Sol`	18	0
`Untuned Gemma`	2	2
`Gemma after SFT · 85 examples, seed 1`	21	20
`Gemma after SFT · 85 examples, seed 2`	19	15

On the 13 validation cases from the same generator, held out of training, the cases passing every blocking check were 3 before training and 2 and 4 after: no consistent improvement. The evaluation set had a higher share of Critical items than the training data (43% vs 18%), and supplier questions revealing stockout durations were concentrated in cases with Critical items. Model size and data volume were not treated as the sole explanation.

Production decision: gpt-6.1-sol stays. Quality was compared on whether the first answer passed every blocking check. The FP8 footprint (13.3 GiB) fits within one L4's memory, but the open-weight model fell short of gpt-6.1-sol (24/25) both before SFT (14/25) and after it (3–10/25), so it did not replace it.

Future work: rejection-sampling fine-tuning is being considered: training only on reasoning-mode answers that pass the production checks, with their reasoning, to reduce both figure errors after changes and internal-action wording in supplier questions.

TECH STACK Python · FastAPI · LangGraph · LangSmith · OpenAI and Anthropic APIs · ChromaDB · MCP · QLoRA SFT · vLLM · Docker · GitHub Actions · Cloud Run